KERNEL OF THE Q FUNCTOR IS THE PARASITIC SUBCATEGORY; Q IS EXACT AND BICONTINUOUS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Stack presentation by a smooth groupoid. Algebraic Stacks, Lemma algebraic-lemma-stack-presentation. Provides X = [U/R] with U a scheme.
- Colimits from coproducts and coequalizers. Categories, Lemma categories-lemma-colimits-coproducts-coequalizers. Reduces colimit commutation to direct sums for an exact functor.
- Parasitic modules form a Serre subcategory. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-parasitic (2)(b). Named inside the proof for the snake-lemma step.

Source reference. Original lemma 0778: sources/stacks-chow/tags.txt, source lines 9193–9193. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 002P: sources/stacks-chow/tags.txt, source lines 106–106. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let $Q : LQCoh^{fbc}(\mathcal{O}_\mathcal{X}) \to QCoh(\mathcal{O}_\mathcal{X})$ be the functor constructed in Lemma 0778.*

1. *The kernel of Q is exactly the collection of parasitic objects of $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$.*
2. *For any object $\mathcal{F}$ of $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ both the kernel and the cokernel of the adjunction map $Q(\mathcal{F}) \to \mathcal{F}$ are parasitic.*
3. *The functor Q is exact and commutes with all limits and colimits.*

Proof. Write $\mathcal{X} = [U/R]$ as in the proof of Lemma 0778. Let $\mathcal{F}$ be an object of $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$. It is clear from the proof of Lemma 0778 that $\mathcal{F}$ is in the kernel of Q if and only if $\mathcal{F}|_{U_{étale}} = 0$. In particular, if $\mathcal{F}$ is parasitic then $\mathcal{F}$ is in the kernel. Next, let $x : V \to \mathcal{X}$ be a flat morphism, where V is a scheme. Set $W = V \times_\mathcal{X} U$ and consider the diagram

$$\begin{array}{ccc} W & \xrightarrow[q]{} & V \\ {\scriptstyle p}\downarrow & & \downarrow \\ U & \longrightarrow & \mathcal{X} \end{array}$$

Note that the projection $p : W \to U$ is flat and the projection $q : W \to V$ is smooth and surjective. This implies that q^*_{small} is a faithful functor on quasi-coherent modules. By assumption $\mathcal{F}$ has the flat base change property so that we obtain

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

$p_{small}^*\mathcal{F}|_{U_{étale}} \cong q_{small}^*\mathcal{F}|_{V_{étale}}$. Thus if $\mathcal{F}$ is in the kernel of Q, then $\mathcal{F}|_{V_{étale}} = 0$ which completes the proof of (1).

Part (2) follows from the discussion above and the fact that the map $Q(\mathcal{F}) \to \mathcal{F}$ becomes an isomorphism after restricting to $U_{étale}$.

To see part (3) note that Q is left exact as a right adjoint. Let $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ be a short exact sequence in $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$. Consider the following commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Q(\mathcal{F}) & \longrightarrow & Q(\mathcal{G}) & \longrightarrow & Q(\mathcal{H}) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle a} & & \downarrow{\scriptstyle b} & & \downarrow{\scriptstyle c} & & \\ 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{G} & \longrightarrow & \mathcal{H} & \longrightarrow & 0 \end{array}$$

Since the kernels and cokernels of a, b, and c are parasitic by part (2) and since the bottom row is a short exact sequence, we see that the top row as a complex of $\mathcal{O}_\mathcal{X}$-modules has parasitic cohomology sheaves (details omitted; this uses that the category of parasitic modules is a Serre subcategory of the category of all modules). By left exactness of Q we see that only exactness at $Q(\mathcal{H})$ is at issue. However, the cokernel $\mathcal{Q}$ of $Q(\mathcal{G}) \to Q(\mathcal{H}))$ may be computed either in $Mod(\mathcal{O}_\mathcal{X})$ or in $QCoh(\mathcal{O}_\mathcal{X})$ with the same result because the inclusion functor $QCoh(\mathcal{O}_\mathcal{X}) \to LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ is a left adjoint and hence right exact. Hence $\mathcal{Q} = Q(\mathcal{Q})$ is both quasi-coherent and parasitic, whence 0 by part (1) as desired.

As a right adjoint Q commutes with all limits. Since Q is exact, to show that Q commutes with all colimits it suffices to show that Q commutes with direct sums, see Categories, Lemma 002P. Let $\mathcal{F}_i$, $i \in I$ be a family of objects of $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$. To see that $Q(\bigoplus \mathcal{F}_i)$ is equal to $\bigoplus Q(\mathcal{F}_i)$ we look at the construction of Q in the proof of Lemma 0778. This uses a presentation $\mathcal{X} = [U/R]$ where U is a scheme. Then $Q(\mathcal{F})$ is computed by first taking the pair $(\mathcal{F}|_{U_{étale}}, \alpha)$ in $QCoh(U, R, s, t, c)$ and then using the equivalence $QCoh(U, R, s, t, c) \cong QCoh(\mathcal{O}_\mathcal{X})$. Since the restriction functor $Mod(\mathcal{O}_\mathcal{X}) \to Mod(\mathcal{O}_{U_{étale}})$, $\mathcal{F} \mapsto \mathcal{F}|_{U_{étale}}$ commutes with direct sums, the desired equality is clear. $\square$